

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 1

Adaptive Classified

Chapter 03 · Simultaneous Equations

With Answers

Questions with full worked solutions

- 14 pages
- Past papers up to June 2026

Simultaneous Equations

Your classified route

5
questions

31
marks

2
families

4
variants

DR ESLAM AHMED

CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

Simultaneous Equations

Families and variants

F01 • Model-parameter Simultaneous Systems

V01 Two-observation Calibration and Prediction

F02 • Nonlinear Systems by Substitution

V01 Linear plus Quadratic System

V02 Polynomial Pair to a Biquadratic

V03 Rational-cubic Intersections with Follow-on Geometry

DR ESLAM AHMED

Model-parameter Simultaneous Systems

WHAT THIS FAMILY ASKS

Use two observed data points to form and solve a linear simultaneous system in two model constants, then invert the calibrated model for a prediction.

VARIANT ROUTE

V01 Two-observation Calibration and Prediction

2 Complete Classified questions

11 total marks

2. A tree was planted.

Exactly 3 years after it was planted, the height of the tree was 2 m.

Exactly 5 years after it was planted, the height of the tree was 2.4 m.

Given that the height, H metres, of the tree, t years after it was planted, can be modelled by the equation

$$H^3 = pt^2 + q$$

where p and q are constants,

(a) find, to 3 significant figures where necessary, the value of p and the value of q . (4)

Exactly T years after the tree was planted, its height was 5 m.

(b) Find the value of T according to the model, giving your answer to one decimal place. (2)

(Total 6 marks)

Question 1

WMA11/01 January 2021

T03 Simultaneous Equations

F01 Model-parameter Simul-
taneous SystemsV01 Two-observation Calibra-
tion and Prediction

1

Substitute the two given heights

When $t = 3$ and $H = 2$, $2^3 = 9p + q$, so $8 = 9p + q$. When $t = 5$ and $H = 2.4$, $2.4^3 = 25p + q$, so $13.824 = 25p + q$.

2

Solve for p and q

Subtracting the two equations gives $16p = 5.824$, so $p = 0.364$. Then $8 = 9(0.364) + q$, giving $q = 4.724$, so $q = 4.72$ to 3 significant figures.

3

Use $H = 5$

For $H = 5$, $125 = 0.364T^2 + 4.724$.

4

Find T

$T^2 = \frac{125 - 4.724}{0.364} = 330.428 \dots$, so $T = 18.177 \dots$. Therefore $T = 18.2$ years to one decimal place.

FINAL ANSWER

(a) $p = 0.364$, $q = 4.72$. (b) $T = 18.2$ years.

3. The population of a town was monitored.

Exactly 5 years after monitoring began, the population was 58 000

Exactly 10 years after monitoring began, the population was 65 000

Given that the population of the town, P thousand, t years after monitoring began can be modelled by the equation

$$P^2 = a + bt^3$$

where a and b are constants,

(a) find the value of a and the value of b .

(3)

According to the model, exactly T years after monitoring began, the population was 85 000

Making your method clear,

(b) find the value of T , giving your answer to one decimal place.

(Solutions relying entirely on calculator technology are not acceptable.)

(2)

(Total for Question 3 is 5 marks)

Question 2

WMA11/01 January 2025

T03 Simultaneous Equations

F01 Model-parameter Simultaneous Systems

V01 Two-observation Calibration and Prediction

1

Use t equals 5The population is 58 thousand, so $58^2 = a + b(5^3)$, giving $3364 = a + 125b$.

2

Use t equals 10The population is 65 thousand, so $65^2 = a + b(10^3)$, giving $4225 = a + 1000b$.

3

Solve for bSubtracting gives $861 = 875b$, so $b = \frac{861}{875} = \frac{123}{125}$.

4

Solve for a

$$a = 3364 - 125 \left(\frac{123}{125} \right) = 3241.$$

5

Use population 85 thousand

$$85^2 = 3241 + \frac{123}{125}T^3.$$

6

Find T

$$T^3 = \frac{125(7225 - 3241)}{123}, \text{ so } T \approx 15.938 \text{ and } T = 15.9 \text{ to one decimal place.}$$

FINAL ANSWER

$$(a) a = 3241, b = \frac{123}{125}. \quad (b) T = 15.9 \text{ years.}$$

Nonlinear Systems by Substitution

WHAT THIS FAMILY ASKS

Isolate one variable or equate two curve expressions, reduce the system to a solvable quadratic structure, and back-substitute to obtain matched ordered pairs and any requested geometry.

VARIANT ROUTE

V01 Linear plus Quadratic System

V02 Polynomial Pair to a Biquadratic

V03 Rational-cubic Intersections with Follow-on Geometry

3 Complete Classified questions

20 total marks

2.

**In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.**

(a) Given

$$y - x = 4$$

and

$$2x^2 - xy = 8$$

show that

$$x^2 - 4x - 8 = 0 \quad (2)$$

(b) Hence solve the simultaneous equations

$$y - x = 4$$

$$2x^2 - xy = 8$$

writing the answers in the form $p + q\sqrt{3}$, where p and q are integers to be found. (4)

(Total for Question 2 is 6 marks)

Question 3

WMA11/01 January 2026

T03 Simultaneous Equations F02 Nonlinear Systems by V01 Linear plus Quadratic Substitution System

1

Use the linear equation

$$y - x = 4, \text{ so } y = x + 4.$$

2

Substitute into the second equation

$$2x^2 - xy = 8 \text{ becomes } 2x^2 - x(x + 4) = 8.$$

3

Show the quadratic

$$2x^2 - x^2 - 4x = 8, \text{ so } x^2 - 4x - 8 = 0.$$

4

Complete the square

$$x^2 - 4x - 8 = 0 \text{ gives } (x - 2)^2 = 12.$$

5

Find x

$$x = 2 \pm \sqrt{12} = 2 \pm 2\sqrt{3}.$$

6

Find y

$$\text{Since } y = x + 4, y = 6 \pm 2\sqrt{3} \text{ with matching signs.}$$

7

State both solutions

$$(x, y) = (2 + 2\sqrt{3}, 6 + 2\sqrt{3}) \text{ or } (2 - 2\sqrt{3}, 6 - 2\sqrt{3}).$$

FINAL ANSWER

$$(x, y) = (2 + 2\sqrt{3}, 6 + 2\sqrt{3}) \text{ or } (2 - 2\sqrt{3}, 6 - 2\sqrt{3}).$$

6. In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

(a) Given that

$$2xy - 3x^2 = 50$$

and

$$y - x^3 + 6x = 0$$

show that

$$2x^4 - 15x^2 - 50 = 0 \quad (2)$$

(b) Hence solve the simultaneous equations

$$2xy - 3x^2 = 50$$

$$y - x^3 + 6x = 0$$

Give your answers in fully simplified surd form. (5)

(Total 7 marks)

Question 4

WMA11/01 May/June 2022

T03 Simultaneous Equations

F02 Nonlinear Systems by
SubstitutionV02 Polynomial Pair to a Bi-
quadratic

1

Rearrange the second equationFrom $y - x^3 + 6x = 0$, we get $y = x^3 - 6x$.

2

Substitute into the first equation $2x(x^3 - 6x) - 3x^2 = 50$, so $2x^4 - 15x^2 - 50 = 0$.

3

Solve the quadratic in x squaredLet $u = x^2$. Then $2u^2 - 15u - 50 = 0$, giving $u = 10$ or $u = -\frac{5}{2}$.

4

Reject the negative valueSince x^2 cannot be negative, $x^2 = 10$, so $x = \pm\sqrt{10}$.

5

Find y $y = x^3 - 6x = x(x^2 - 6) = x(10 - 6) = 4x$. Therefore $y = 4\sqrt{10}$ when $x = \sqrt{10}$ and $y = -4\sqrt{10}$ when $x = -\sqrt{10}$.**FINAL ANSWER** $(x, y) = (\sqrt{10}, 4\sqrt{10})$ or $(-\sqrt{10}, -4\sqrt{10})$.

8.

In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

The curve C_1 has equation

$$xy = \frac{15}{2} - 5x \quad x \neq 0$$

The curve C_2 has equation

$$y = x^3 - \frac{7}{2}x - 5$$

(a) Show that C_1 and C_2 meet when

$$2x^4 - 7x^2 - 15 = 0 \quad (2)$$

Given that C_1 and C_2 meet at points P and Q

(b) find, using algebra, the exact distance PQ (5)

(Total for Question 8 is 7 marks)

Question 5

WMA11/01 October 2023

T03 Simultaneous Equations

F02 Nonlinear Systems by
SubstitutionV03 Rational-cubic Intersec-
tions with Follow-on Geometry

1

Write C1 in terms of y

$$xy = \frac{15}{2} - 5x, \text{ so } y = \frac{15}{2x} - 5.$$

2

Equate the two curves

$$\frac{15}{2x} - 5 = x^3 - \frac{7}{2}x - 5.$$

3

Multiply by 2x

$$15 = 2x^4 - 7x^2, \text{ hence } 2x^4 - 7x^2 - 15 = 0.$$

4

Solve for x

$$\text{Let } u = x^2. \text{ Then } 2u^2 - 7u - 15 = 0 = (2u + 3)(u - 5), \text{ so } x^2 = 5 \text{ and } x = \pm\sqrt{5}.$$

5

Find the y-coordinates

$$\text{Using } y = \frac{15}{2x} - 5, \text{ the points are } \left(\sqrt{5}, \frac{3\sqrt{5}}{2} - 5\right) \text{ and } \left(-\sqrt{5}, -\frac{3\sqrt{5}}{2} - 5\right).$$

6

Use the distance formula

$$\Delta x = 2\sqrt{5} \text{ and } \Delta y = 3\sqrt{5}, \text{ so } PQ = \sqrt{(2\sqrt{5})^2 + (3\sqrt{5})^2} = \sqrt{65}.$$

FINAL ANSWER

$$(a) 2x^4 - 7x^2 - 15 = 0. \quad (b) PQ = \sqrt{65}.$$